select earphone with a sensitivity of 100 to 110dB/mW. You will be able to listen to the radio with the earphones.

Circuit 4 without antenna and earth

You can make a crystal radio work even without an outdoor antenna and earth as shown in Fig. 14. This is an interesting circuit for those who do not have adequate space for erecting a long antenna, or do not wish to use an antenna and earth for some reason and still receive the stations.

The antenna used in this circuit is called 'Loop Antenna.' It is made of eight turns of a multi-strand hook-up wire (14/36) on a 120cm × 120cm (4 × 4 feet) on a wooden frame with legs to make it stand on the floor. You can also use smaller frames but bigger the size, better the signal output. The photo in Fig. 12 shows the crystal radio using the loop antenna made by the authors.

Components required:

1. Wooden frame of about 120cm x 120cm with legs
2. Multistrand hookup wire (14/36) – 40 metres
3. Variable capacitor rated 270pF (dual gang) – 1
4. 1N34A diode – 1, or any germanium diode with Vf less than 300mV
5. Perforated PC board of 7.5cm × 3.8cm (3" × 1.5") – 1

Make a wooden frame of about 120cm × 120cm with legs so that it can stand on the floor. Nail the four corners of the frame with 10cm iron nails or screws. Start running the hook-up wire with a knot from the left top nail, leaving about 15 cms of wire at the start. Make eight turns clockwise from left top, right top, right bottom, left bottom, and back to left top nail, and leave an extra 15cm of wire for making connections with PC board circuit. Maintain a small gap of 4 to 5mm between each turn.

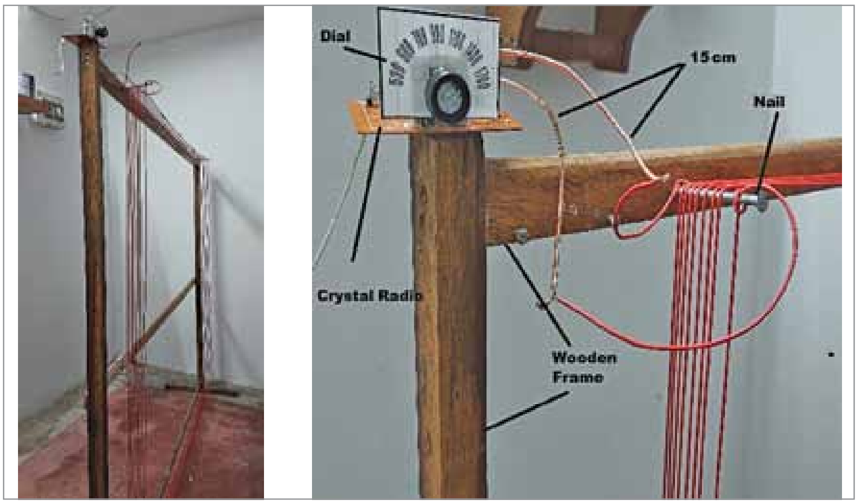

Fig. 12: Making of indoor loop antenna

Connect the loop, variable capacitor, and diode as in Circuit 4. The terminals to be connected to the headphone may be connected with miniature crocodile clips for easy interchange between high-impedance headphones or the primary of the matching transformer. Tune the variable capacitor either clockwise or anticlockwise, slowly, and you will hear the audio when the frequency of the local station matches with the capacitance. The loop antenna is very directional and hence it should be rotated slowly and oriented towards the direction of maximum volume.

The advantage of using a loop antenna in crystal radio is that it does not require an external antenna or earth. The radio signal picked up by the loop antenna is slightly lower than that from an external antenna.

Troubleshooting

Crystal radios use only an antenna to receive the radio signal and a diode to rectify/convert the radio signal to audio signal to be heard using a headphone. It is entirely passive and has no active components like transistors or ICs, and no battery or electricity is used. Hence the volume obtained in the headphone/earphone is relatively low. To receive a radio station properly, check the following points:

- You have a long wire antenna of a minimum of 18 metres at a height of 6 to 9 metres. If you live in a city where the power of the radio station is 200 or 300 kilowatts, 18 metres long antenna gives the best results. If you live in low-power area, make the antenna longer.
- You have a good earth connection.

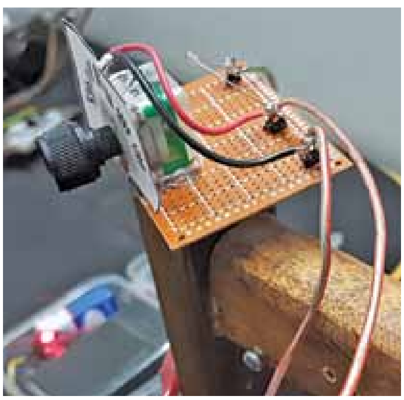

Fig. 13: Author's prototype

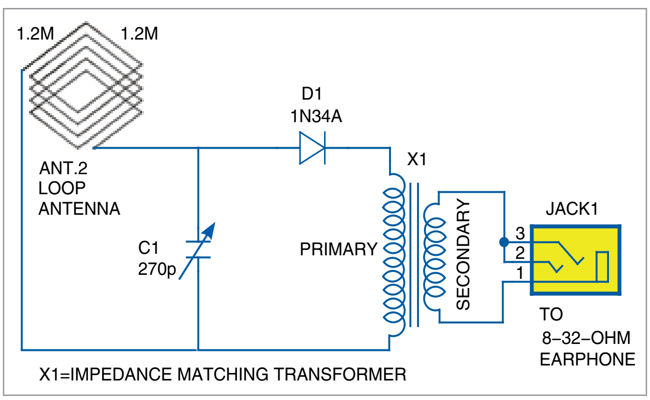

Fig. 14: Circuit 4 connections

- A diode is the heart of the crystal radio. Confirm that you have chosen the right type of germanium diode. Variations are found in the diodes available in the market, as most of the diodes sold as 1N34A have no marking on them and you cannot be sure whether you are using a correct